

II.1. Risk Factors for Squamous Cell Carcinoma (SCC)

II.1.1. Environmental and Exogenous Factors

The occurrence of SCC is associated with total cumulative lifetime UV dose. The most commonly affected sites include the face, back of the hands, and forearms. In a Spanish study, more than 92% of SCC cases occurred in these areas. UVB (290–320 nm) and UVA (320–400 nm) both play a role in carcinogenesis. For most SCCs, UV-induced mutations are observed in the *P53* gene.

Artificial sources of UV radiation have also been implicated. PUVA therapy, particularly when exceeding 200 sessions, is associated with an increased risk of SCC. Tanning salon UV sources are not harmless and must be considered among other risk factors. According to the 2005 report by the French Agency for Environmental and Occupational Health Safety (AFSSE), "Evaluation of risks associated with exposure to UV radiation," the risk of skin cancer (carcinoma or melanoma) increases by a factor of 1.10 with 30 tanning sessions per year over a 10-year period, and by 1.39 with 100 sessions.

Other exogenous risk factors include arsenic, pesticides, hydrocarbons, tobacco (particularly for SCC of the lower lip), ionizing radiation, and prolonged local chemotherapy.

II.1.2. Constitutional Factors

The main constitutional factor is skin phototype, which is genetically determined. Individuals with a poor capacity for tanning are at higher risk.

In xeroderma pigmentosum—a hereditary recessive disorder—deficiency in DNA repair enzymes leads to an inability to correct photoinduced DNA

damage, resulting in a markedly increased risk of SCC, actinic keratosis (AK), and keratoacanthoma.

II.1.3. Other Risk Factors

The following factors contribute to less than 1% of SCC cases: iatrogenic immunosuppression, chronic inflammation, chronic leg ulcers, and scarring.

Human papillomavirus (HPV) infections have been implicated, particularly in genital and anal SCC. HPV is also found in extragenital sites, in both sun-exposed and non-exposed areas, and in both immunocompromised and immunocompetent individuals. However, HPV alone is insufficient to induce malignant transformation. In organ transplant recipients, three synergistic risk factors are present: UV exposure, immunosuppression, and HPV infection.

Key Points

- Squamous cell carcinoma (SCC) encompasses a group of malignant primary epithelial skin tumors with malpighian differentiation and is distinct from basal cell carcinoma (BCC).
- The primary factor promoting the development of SCC, Bowen's disease (BD), or actinic keratosis (AK) is total lifetime UV exposure, whether from natural sunlight or artificial sources.
- Skin phototype, a genetically determined trait, influences the risk of developing AK, BD, or SCC.
- Several co-carcinogens have been identified, some site-specific—such as tobacco for actinic cheilitis and SCC of the lower lip, and HPV for genital or anal SCC.
- Certain medical conditions predispose to SCC, including chronic inflammation, chronic wounds, and immunosuppression.

II.2. Precursors of SCC

II.2.1. Actinic Keratosis

II.2.1.1. Clinical, Histological, Epidemiological Aspects and Natural History

Clinical Aspects

Actinic keratosis (AK) consists of common lesions, especially in fair-skinned individuals, occurring in sun-exposed areas such as the face, dorsal aspects of the hands, and scalp in balding individuals. These lesions are often accompanied by other signs of photoaging, such as wrinkles and freckles.

Diagnosis is primarily clinical, based on the presence of flat, rough lesions of variable thickness, sometimes more visible than palpable. Lesions typically measure 1 cm or less in diameter and exhibit varying degrees of erythema. They may be pigmented and are often covered with a dry, adherent, yellowish or brownish keratin layer.

In clinical practice, isolated or few AKs are distinguished from multiple AKs, which may coalesce into plaques—particularly on the scalp.

On the lower lip, actinic cheilitis represents the labial counterpart of AK. This condition is uniquely associated with tobacco smoking as a secondary carcinogenic factor and carries a higher risk of metastatic SCC compared to other sites.